AquaRun: Bottle Swap Model Changes

Summary of changes · Returnable products only · October 2026

Goal

Make the bottle exchange (swap) model explicit and apply it only to returnable products (`products.is_returnable = true`). Non-returnable items are sold outright and ignored by the bottle ledger and balance. The 8-status order flow and the existing RLS policies were not changed.

Files changed

Step	File	Change
A	`js/services/productsService.js`	Selects `is_returnable`. `saveProduct` accepts `isReturnable`. On update it only writes the field when it is passed, so other callers never overwrite it. New products default to `true`.
A	`js/pages/admin/products.js`	Checkbox "Customer returns the empty container" in Add/Edit product (checked by default). Each product card shows a "Returnable" or "Sold outright" pill.
B	`js/services/ledgerService.js`	New `getOrderReturnableQty(orderId)`, which throws if the view returns no row, so permission problems are visible. New `getCustomerBalanceForOrder(orderId)`, which calls the RPC.
B	`js/pages/rider/complete-delivery.js`	"Bottles delivered" is prefilled with `returnable_qty`. "Empties collected" starts at 0. Read-only hint "Expected empties: N", where N = min(returnable_qty, balance before delivery). A warning shows when empties are below expected, but it never blocks. If returnable_qty is 0, it shows "No returnable items in this order" and both counts stay at 0.
B	`css/pages/rider/complete-delivery.css`	Moved the warning spacing from an inline style into a CSS class rule.
C	`sql/07_ready_uses_returnable.sql` (new)	Grants select on `order_returnable_qty`. Replaces `mark_ready` so it sets bottles refilled from `returnable_qty`. Adds `get_customer_balance_for_order(uuid)` (security definer; only the assigned rider, staff or admin can call it). `04_functions.sql` was not edited.
D	`js/pages/customer/new-order.js`	One cart notice, shown only when the cart has a returnable item. Balance > 0: "Please have your empty gallons ready..." No earlier deliveries: "First order: no empty gallons needed." It uses only `customer_bottle_balance` and the customer's own orders. If a lookup fails, no notice is shown.

E	`js/pages/customer/bottle-balance.js`	Added the line "This is the number of gallons you hold. Hand empties to the rider on your next delivery." The existing return reminder is kept.
F	Sysplan (Word)	Skipped. You will update this file outside the repo.

Rules implemented

- Only returnable items count toward bottles refilled, bottles delivered, expected empties and the balance.
- First-time customer: expected empties is 0, so the balance rises by the number of gallons delivered.
- Regular customer: expected empties equals the gallons delivered, capped by the current balance.
- Customer short on empties: the rider records the actual count, a warning shows, and the difference stays on the balance.
- Balance formula is unchanged: bottles delivered minus empties collected.
- Orders with only non-returnable items: 0 expected, 0 counted, balance unchanged, and all 8 statuses still apply.

Checks performed

- `products.is_returnable` and `order_returnable_qty` exist, and their column names match `order_items`.
- `bottle_ledger.order_id` has a unique constraint, so `on conflict (order_id)` in `mark_ready` is valid.
- `record_delivery` accepts delivered = 0 and empties = 0 (it only rejects null or negative values). No change was needed.
- `deliveryService.js` passes counts through and never uses the full order quantity.

Still to do: Run `sql/07_ready_uses_returnable.sql` in the Supabase SQL Editor. It was not applied from here. The 7 test cases from the brief were checked against the code logic, but they have not been run end-to-end against the live database yet.